ENTERPRISE UML 2.5 SPECIFICATION

Eventura Architecture Specification

Complete 14 Structural & Behavioral UML Diagrams

Formal software engineering architecture documentation for **Eventura (powered by Clientura)**. This definitive specification models the multi-tenant architecture, security boundaries, domain entities, and asynchronous runtime workflows across all 14 standard UML 2.5 diagram types.

Table of Architecture Diagrams (14 Total)

PART I: STRUCTURAL DIAGRAMS

01. Class Diagram (Core Domain Entities)

02. Object Diagram (Runtime Event Snapshot)

03. Component Diagram (Internal Modules & Ports)

04. Composite Structure Diagram (Classifier Parts)

05. Deployment Diagram (Physical Nodes & Tiers)

06. Package Diagram (Layered Namespace Modules)

07. Profile Diagram (UML Extensions & Stereotypes)

PART II: BEHAVIORAL DIAGRAMS

08. Use Case Diagram (Stakeholders & Operations)

09. Activity Diagram (End-to-End Event Workflow)

10. State Machine Diagram (Event & Submission States)

11. Sequence Diagram (Submission & Judging Pipeline)

12. Communication Diagram (Collaboration Topology)

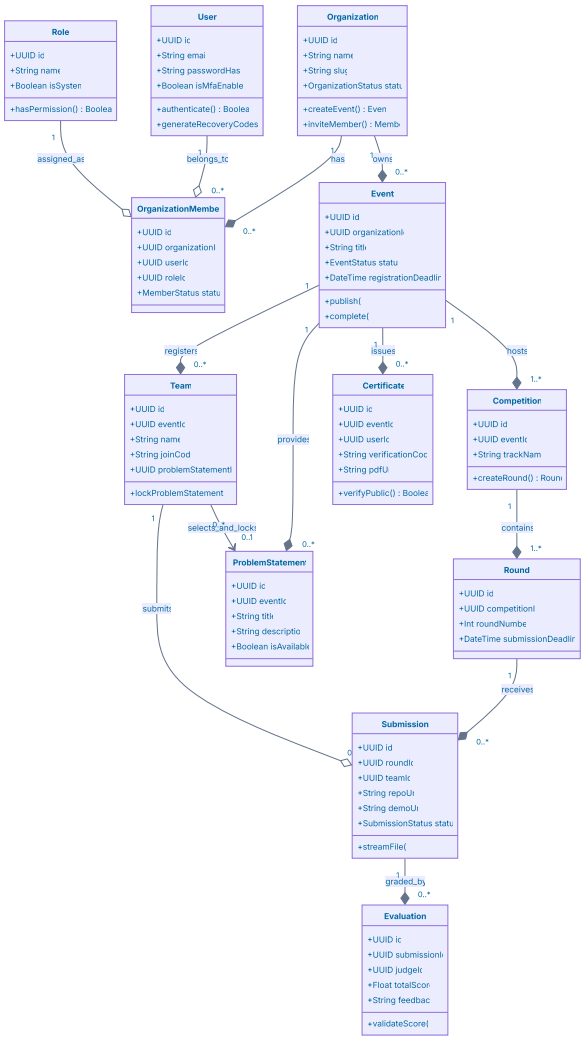
13. Timing Diagram (Linear Lifecycle State Transitions)

14. Interaction Overview Diagram (Process Framing)

STRUCTURAL DIAGRAM 01 OF 14

Class Diagram: Core Domain Entities & Relationships

UML 2.5 Structural Domain Model



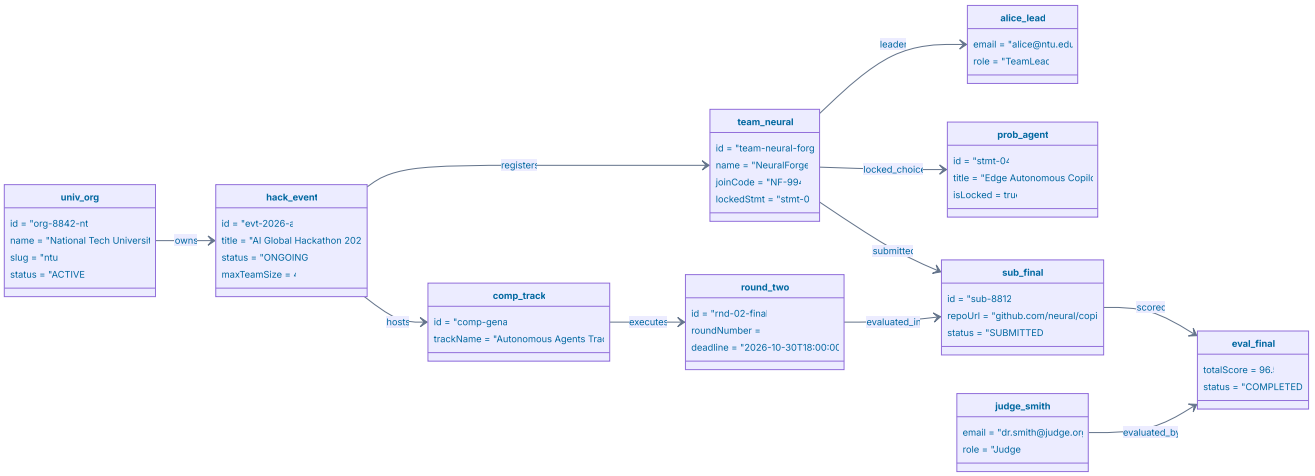
ARCHITECTURAL SPECIFICATION & CONTEXT

The Class Diagram models the core entity-relational schema of Eventura. It captures the multi-tenant container (`Organization`), RBAC mappings (`User`, `OrganizationMember`, `Role`), competition lifecycle, permanent problem statement locking, submission tracking, judge rubric evaluations, and verifiable certificates.

Composite Aggregations | Multi-Tenant Scoping | ACID Relational Mappings | Multiplicity Constraints

Object Diagram: Runtime Event Snapshot

UML 2.5 Structural
Instance Graph



ARCHITECTURAL SPECIFICATION & CONTEXT

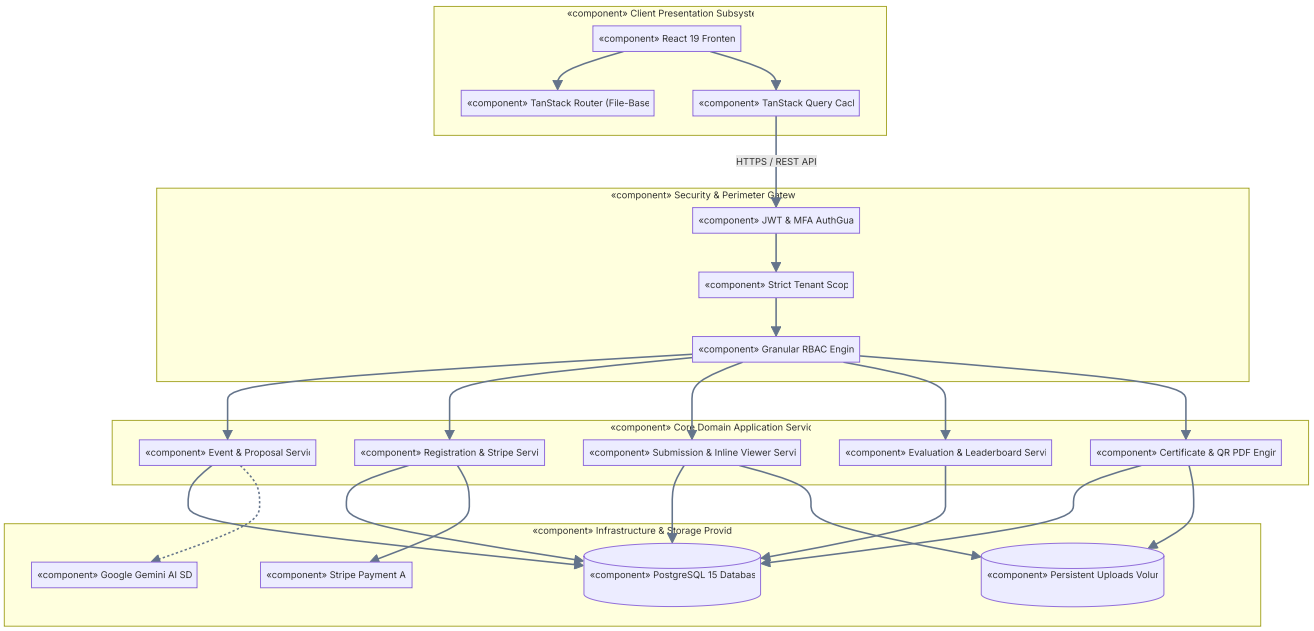
The Object Diagram illustrates a live, concrete runtime state during an ongoing competition. It validates the instantiation of `National Tech University` holding an active hackathon, team `NeuralForge` locking problem statement `Edge Autonomous Copilot`, submitting code for Round 2, and receiving an official score of 96.5 from `Dr. Smith`.

Concrete Runtime Snapshot | Attribute Value Bindings | Live State Validation | Instance Associations

STRUCTURAL DIAGRAM 03 OF 14

Component Diagram: Subsystems, Interfaces & Ports

UML 2.5 Structural
Component Topology



ARCHITECTURAL SPECIFICATION & CONTEXT

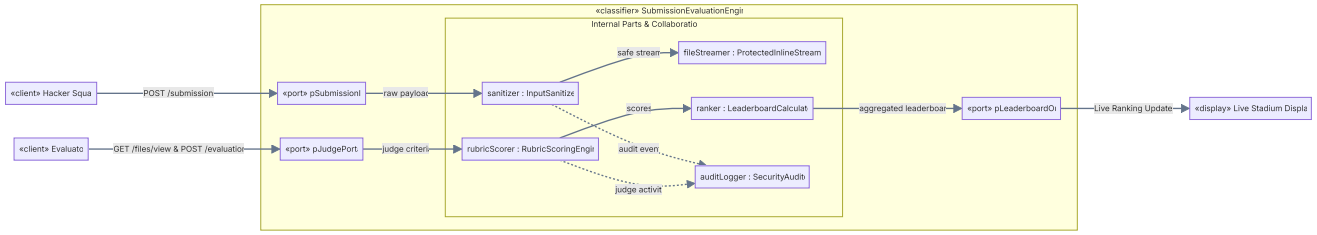
The Component Diagram details the modular composition and provided/required interface boundaries of Eventura. The client presentation subsystem communicates exclusively through the security perimeter gateway, which guards the core domain application services before persisting records into PostgreSQL, local/cloud storage, or external APIs.

Modular Decoupling | Perimeter Gateway Ports | External Service Adapters | Infrastructure Boundaries

STRUCTURAL DIAGRAM 04 OF 14

Composite Structure Diagram: Submission & Evaluation Classifier

UML 2.5 Structural
Internal Classifier
Architecture

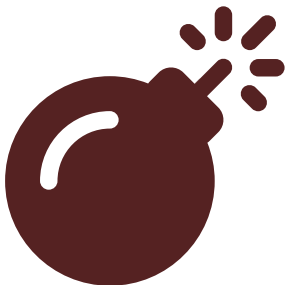


ARCHITECTURAL SPECIFICATION & CONTEXT

The Composite Structure Diagram zooms into the internal wiring of the `SubmissionEvaluationEngine` classifier. It reveals the internal ports, sanitized file ingestion, protected inline streaming, rubric calculation parts, and real-time ranker collaboration without exposing internal plumbing to outside consumers.

Internal Classifier Parts | Port-to-Port Connectors | Role Bindings | Sandboxed Stream Processing

Deployment Diagram: Physical Hardware & Cloud Topology

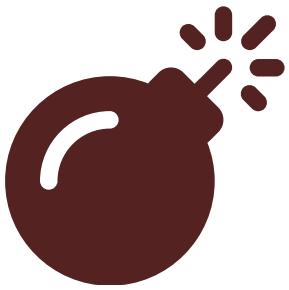


ARCHITECTURAL SPECIFICATION & CONTEXT

The Deployment Diagram specifies the physical and virtual nodes, execution environments, network protocols, and compiled artifact deployments. It illustrates edge TLS termination, NGINX path-based routing dividing traffic between the Nitro SSR server (8080) and Express API (3000), and secured database connections.

Physical & Virtual Nodes | Execution Environments | Artifact Deployment | Network Protocol Bindings

Package Diagram: Codebase Namespaces & Layer Dependencies



ARCHITECTURAL SPECIFICATION & CONTEXT

The Package Diagram articulates the namespace boundaries and unidirectional dependencies across the codebase. It strictly maintains layered architecture principles where routes delegate to controllers, controllers invoke services, services operate on repositories, and repositories interface with Prisma data packages without circular imports.

Unidirectional Flow | Namespace Encapsulation | Layer Decoupling | Circular Dependency Prevention

Profile Diagram: Multi-Tenant Hackathon UML Extension



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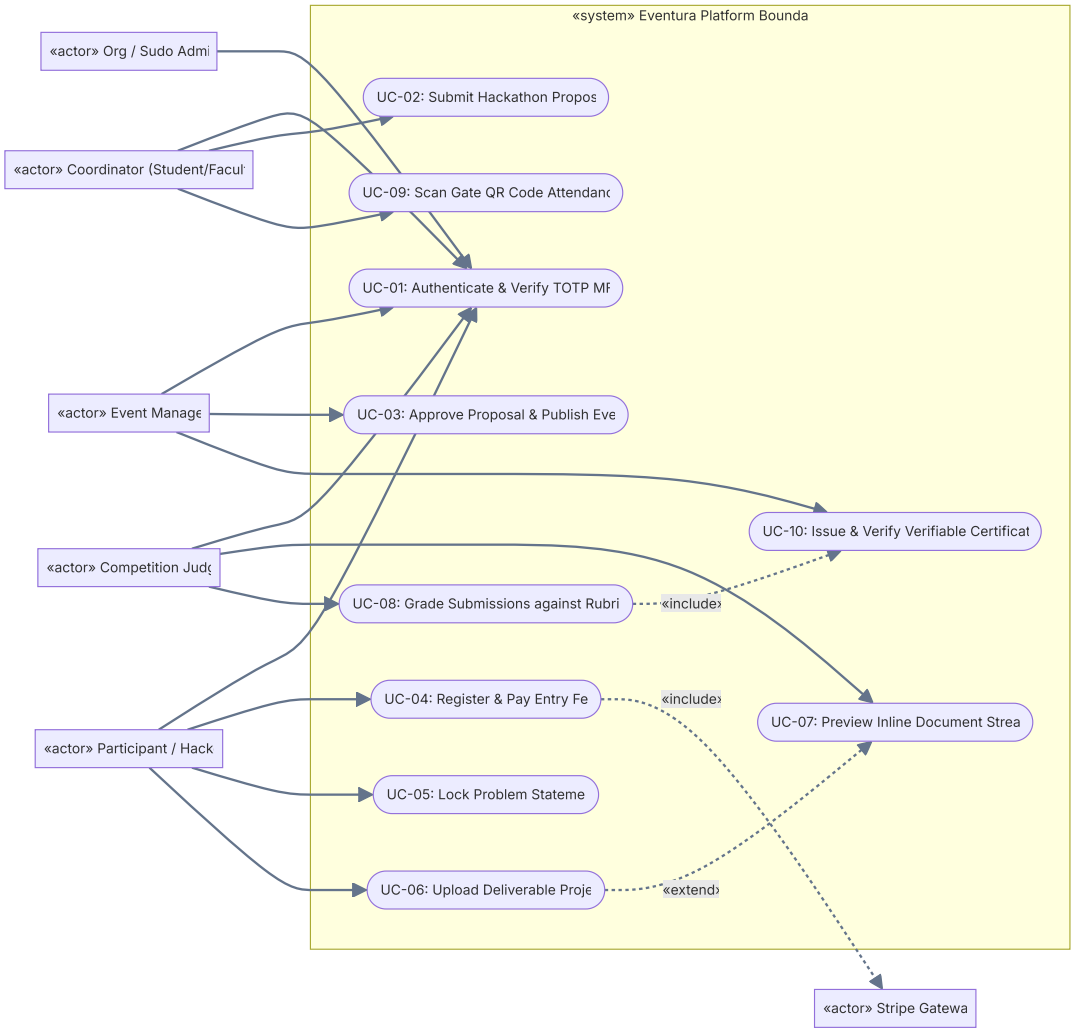
ARCHITECTURAL SPECIFICATION & CONTEXT

The Profile Diagram provides a formal UML extension profile customized for Eventura's domain. It introduces domain-specific stereotypes (`«TenantScoped»`, `«RBACProtected»`, `«AuditLogged»`, `«ImmutableLock»`, and `«CryptographicCredential»`) with custom tagged values extending standard UML metaclasses.

Metamodel Extensions | Custom Stereotypes | Tagged Values | Domain Policy Constraints

Use Case Diagram: Actors & System Capabilities

UML 2.5 Behavioral
Stakeholder Interactions



ARCHITECTURAL SPECIFICATION & CONTEXT

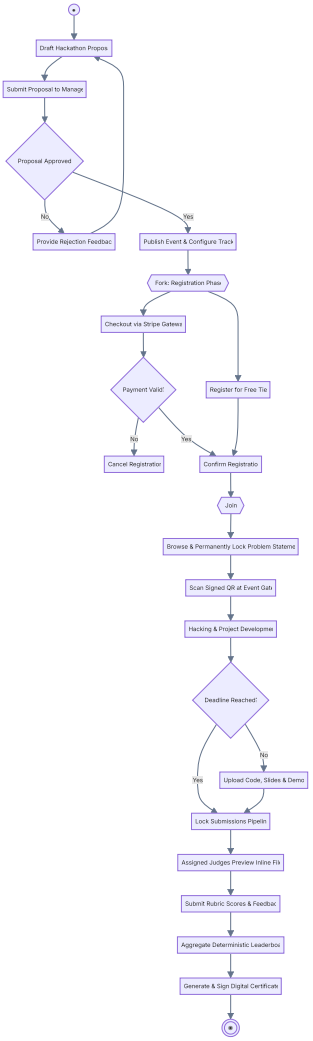
The Use Case Diagram defines the operational interactions between external actors and system boundaries. It models key functional goals from MFA authentication and hackathon proposal approvals, to registration payments, permanent statement locking, inline evaluation preview, gate QR check-ins, and cryptographic certificate issuance.

Actor Role Mapping | System Boundary | «include» & «extend» Relations | External Payment Gateway

BEHAVIORAL DIAGRAM 09 OF 14

Activity Diagram: End-to-End Hackathon Workflow

UML 2.5 Behavioral
Workflow Control Flow

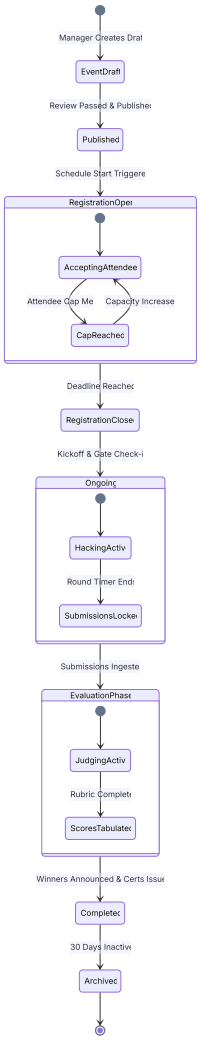


ARCHITECTURAL SPECIFICATION & CONTEXT

The Activity Diagram models the step-by-step logic flow of a hackathon execution. It features fork/join concurrency during free vs paid registration paths, gate condition decision diamonds, deadline gates for deliverable uploads, and sequential score aggregation concluding with automated digital certificate issuance.

Fork & Join Synchronization | Decision Diamonds | Payment Verification Branch | Deadline Constraints

State Machine Diagram: Dual Lifecycle State Automata



ARCHITECTURAL SPECIFICATION & CONTEXT

The State Machine Diagram documents the finite state automata of the `Event` lifecycle. It specifies entry, exit, and transition triggers from initial draft and capacity-bounded registration to ongoing competition hacking, sandboxed evaluation, winner finalization, and cold archival.

Finite State Automata | Nested State Machine | Transition Invariants | Guard Conditions

BEHAVIORAL DIAGRAM 11 OF 14

Sequence Diagram: Submission & Judge Evaluation Pipeline

UML 2.5 Behavioral
Temporal Message
Lifelines



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ARCHITECTURAL SPECIFICATION & CONTEXT

The Sequence Diagram models the temporal message exchange between actors, security filters, controller endpoints, and database transactions. It traces deliverable upload verification, the secure `Content-Disposition: inline` file streaming authorization, and atomic rubric score persistence.

Temporal Message Ordering | Lifeline Activation Bars | Synchronous Request-Response | Atomic Transaction Scope

Communication Diagram: Attendee Registration Topology



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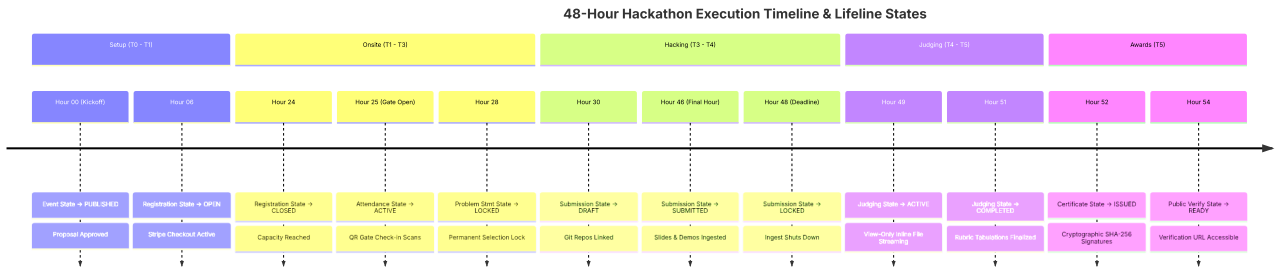
ARCHITECTURAL SPECIFICATION & CONTEXT

The Communication Diagram (formerly Collaboration Diagram) depicts the architectural topology of collaborating runtime objects. It emphasizes spatial object relationships with numbered messages orchestrating user ticket registration, tenant header verification, Stripe checkout session generation, database persistence, and QR attendance signing.

Object Collaboration Links | Numbered Message Sequence | Spatial Relationship Graph | Service Orchestration

Timing Diagram: Linear Competition Timeline & State Transitions

UML 2.5 Behavioral
Time-Constrained
Lifelines

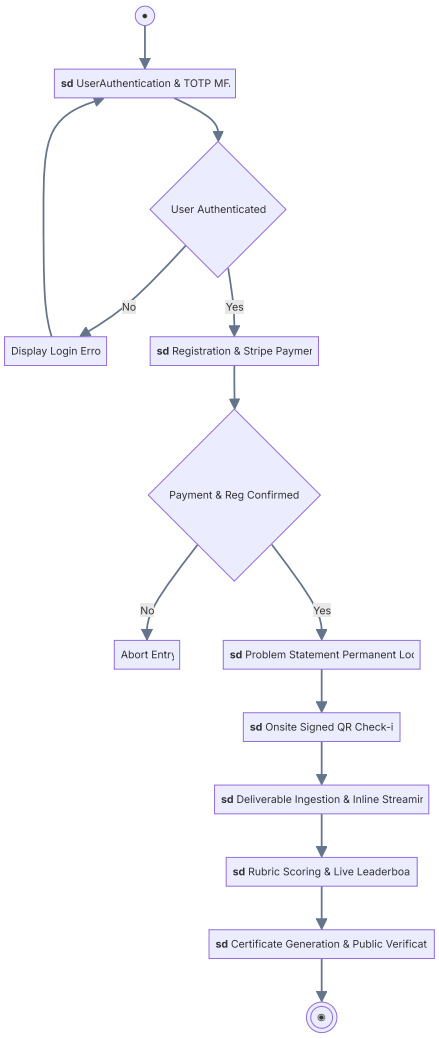


ARCHITECTURAL SPECIFICATION & CONTEXT

The Timing Diagram models the exact state changes of critical lifelines over a continuous 48-hour event timeline. It illustrates strict time constraints: registration closing at Hour 24, permanent statement locking at Hour 28, submission ingestion locking at Hour 48, and cryptographic certificate issuance at Hour 52.

Linear Event Timeline | State Duration Constraints | Synchronized Lifelines | Milestone Deadlines

Interaction Overview Diagram: High-Level Process Framing



ARCHITECTURAL SPECIFICATION & CONTEXT

The Interaction Overview Diagram coordinates high-level business flows by nesting sequence diagram frames within an activity-like decision graph. It structures the complete participant journey through sequential frames: `sd UserAuthentication`, `sd Registration`, `sd ProblemStatementLock`, `sd QRCheckin`, `sd Submissions`, and `sd CertificateVerification`.

Sequence Diagram Framing | Decision Invariants | Macro Process Control | End-to-End Orchestration